

Robin Lange

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Robotics and Autonomous Systems | Perception, Mission Logic and Physical Testing

Education

Georgia Southern University **Aug. 2023–May 2027**
B.S. Mechanical Engineering, Vehicle Design & Testing concentration (in progress) *GPA: 4.00/4.00*

Institutional GPA through Spring 2026; President's List in every completed graded term. Honors College Scholar; Tau Beta Pi; Allen E. Paulson Scholar.

Descartes-Gymnasium Neuburg an der Donau **Sept. 2014–June 2023**
Allgemeine Hochschulreife / General Higher Education Entrance Qualification *Germany*

Final German grade: 1.2 (1.0 highest); 791/900 points.

Research Experience

Institute for Human & Machine Cognition (IHMC) **May–Aug. 2026**

National Center for Collaborative Autonomy (NCCA)

Collaborative Robotics Autonomy Intern *Pensacola, FL*

- Built the Rust/Python autonomy codebase from scratch for a collaborative platform with two Boston Dynamics Spot robots; implemented pickup/drop-off sequencing and Joint Activity Graph (JAG) coordination.
- Combined inverse kinematics for coarse approach with local perception for final grasping, enabling successful hardware pickups despite tracking drift and imperfect perception.
- Implemented coordinate transforms and Zenoh communication; used Motive motion capture, MCAP/Foxglove logging and robot SDKs; added tracking markers to a manually operated Husky AGV.

Georgia Southern University **Oct. 2023–present**
Undergraduate Researcher, Intelligent Vehicles Navigation Research Laboratory *Statesboro, GA*

- Integrated visual SLAM, nvblox mapping, Nav2 and behavior-tree mission logic on Jetson/RealSense UAV hardware; evaluated localization in a 110-second cart trial (14.5 cm RMSE on valid reference segments).
- Developed a LiDAR hand-motion interface for UAV control; coauthored two published SAE papers on lane perception and vehicle-simulation calibration.
- Developing a MATLAB/AVL longitudinal-control baseline and simulator validation for an Honors thesis; learned-prediction comparisons are planned.

CARISSMA Institute of Safety in Future Mobility **July 1–Aug. 15, 2025**

Technische Hochschule Ingolstadt

Vehicle Testing Research Intern *Ingolstadt, Germany*

- Designed and built the covering, quick-assembly setup and fold-away mechanisms of a soft truck-rear crash target, developing an earlier CAD/foam concept into a testable physical prototype.
- Researched radar, LiDAR and camera sensing; supported vehicle tests and analyzed qualitative LiDAR/4D-radar returns with team scripts.

Fraunhofer Application Center **June 1–Aug. 11, 2024**

“Connected Mobility and Infrastructure” – Fraunhofer IVI

Autonomous Systems and AI Intern *Ingolstadt, Germany*

- Implemented hybrid state-machine/behavior-tree mission intelligence in Python, ROS 2 and py_trees for ALBACOPTER, translating flight-phase requirements into executable logic.
- Verified emergency interruption across flight phases, evaluated resource use and documented ALBACOPTER simulation results.

Technical Skills

Programming: Python, Rust, MATLAB, C++; functional/procedural programming; entity-component-system (ECS) patterns.

Robotics: ROS 2/DDS, Zenoh, Protocol Buffers; inverse kinematics, behavior trees, coordinate transforms, visual SLAM.

Hardware and tools: Boston Dynamics Spot, motion capture, LiDAR, RGB-D cameras, Jetson, Gazebo, SolidWorks, Git/Linux, MCAP/Foxglove.

Selected Awards and Scholarships

Third Place Undergraduate, Gulfstream Innovation, Engineering and Flight Excellence Awards – April 24, 2026. *Development of a Pilot AI for Autonomous UAV Inspection in Industry 5.0 Using Visual SLAM and Behavior Trees*; CEC Research Symposium.

Second Place Undergraduate, Georgia Power Innovation Awards – April 25, 2025. *Drone Guiding Through Equivalent Hand Movement Transversion*; CEC Research Symposium.

NCUR Engineering Division Video Contest Awardee – Council on Undergraduate Research, 2026.

Honors College Continuing Scholarship (2024–27); Charles Robert Pound, Sr. Scholarship (2026–27); Paulson Scholar (2025); Vickie Hawkins Scholarship (2024).

Published Research

Soloiu, V., Sutton, T., Mehrzed, S., **Lange, R.**, Zimmerman, C., and Peralta Lopez, G. “Data-Driven Calibration of Vehicle Simulations for Accurate Control Response in Development and Testing.” *SAE Technical Paper 2026-01-0058*, 2026. doi:10.4271/2026-01-0058.

Soloiu, V., Mehrzed, S., Kroeger, L., Pierce, K., Sutton, T., and **Lange, R.** “Development of a Rule-Based Lidar Lane-Keeping Assistance System for Autonomous Vehicles Applications.” *SAE Technical Paper 2025-01-8025*, 2025. doi:10.4271/2025-01-8025.

Preprint and Manuscript in Preparation

Soloiu, V., Sutton, T., **Lange, R.**, and Zimmerman, C. “Warehouse Platooning Framework Using a Sensor-Rich Leader and Minimal Perception Follower.” SSRN 7067358. Preprint; not peer reviewed.

Lange, R., Sezgin, F., Inderst, M., Sutton, T., Lugner, R., Brandmeier, T., and Soloiu, V. “Design and Validation of a Soft Crash Target for Multimodal Sensor-Based Truck Underride Testing.” Manuscript in preparation.

Lange, R. “When to Trust Learned Lead-Vehicle Predictions: Reliability-Gated Preview for Safety-Supervised Adaptive Cruise Control.” Honors thesis in progress; journal paper planned.

Additional Research Presentations

Southern Regional Honors Council Conference, 2026: Intelligent UAV Navigation with V-SLAM and Behavior Trees for Industry 5.0 Applications.

Southern Regional Honors Council Conference, March 2025, Chattanooga, TN: 3D-Space Hand Trajectory Tracking for Intuitive UAV Control.

GS4 symposium: Enhancing UAV Command through 3D-LiDAR-Driven Hand Movement Recognition.

Teaching and Academic Preparation

Physics I/II teaching assistant (Jan. 2025–present): evaluate experimental reports. **Computing for Engineers laboratory assistant** (Jan.–May 2024): guided MATLAB and programming labs. **First-Year Experience peer leader** (Aug.–Dec. 2024).

Completed coursework: Calculus I–III, Differential Equations, Numerical Methods, Dynamics, Mechatronics, Circuit Analysis, Program Design, Logic Circuit Design.

Languages: German (native), English (fluent), French (DELF B1).

Activity History Complementing Education on Page 1

Georgia Southern University

Aug. 2023–present

B.S. Mechanical Engineering in progress; May 2027 graduation target

Statesboro, GA

Full-time undergraduate study with concurrent research, teaching and the summer internships listed on pages 1–2.

Wittelsbacher Golfclub

June–Aug. 2023

Administrative and Customer Service Assistant

Neuburg an der Donau, Germany

Additional Employment and Teaching

Physics Teaching Assistant

Jan. 2025–present

Principles of Physics I and II, Georgia Southern University

First-Year Experience Peer Leader

Aug.–Dec. 2024

Georgia Southern University

Laboratory Assistant, Computing for Engineers

Jan.–May 2024

Georgia Southern University

Event Assistant

Dec. 2023

Maritim Hotel & Congress Centrum

Ingolstadt, Germany

Student Assistant

Dec. 2023

Eagle Dining Services

Statesboro, GA

Tutor

Jan.–July 2023

Genius Tutoring and Languages

Neuburg an der Donau, Germany

Mathematics, physics, chemistry, information technology and English.

School work-experience placement

Apr. 19–20, 2022

Airbus Helicopters

Donauwörth, Germany

Exposure to the aircraft-mechanic field.

Short aviation/technical placement

June 7–10, 2022

Airbus Defence and Space

Manching, Germany

Observed aircraft assembly and manufactured model prototypes using workshop equipment.

Earlier Academic Work

Physics W-Seminar paper: “Der Spin des Golfballs.” Descartes-Gymnasium Neuburg an der Donau.